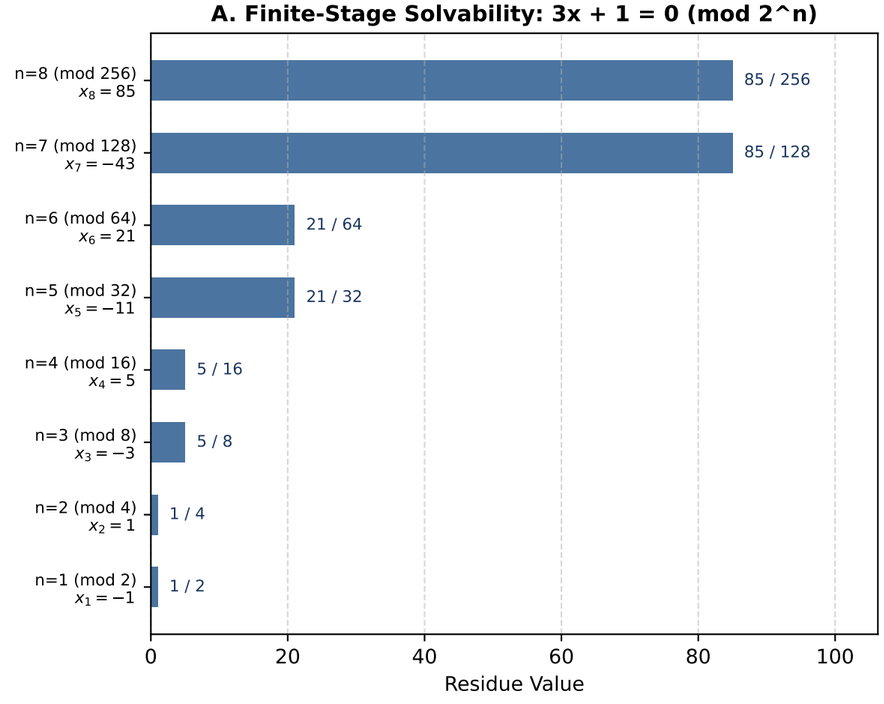
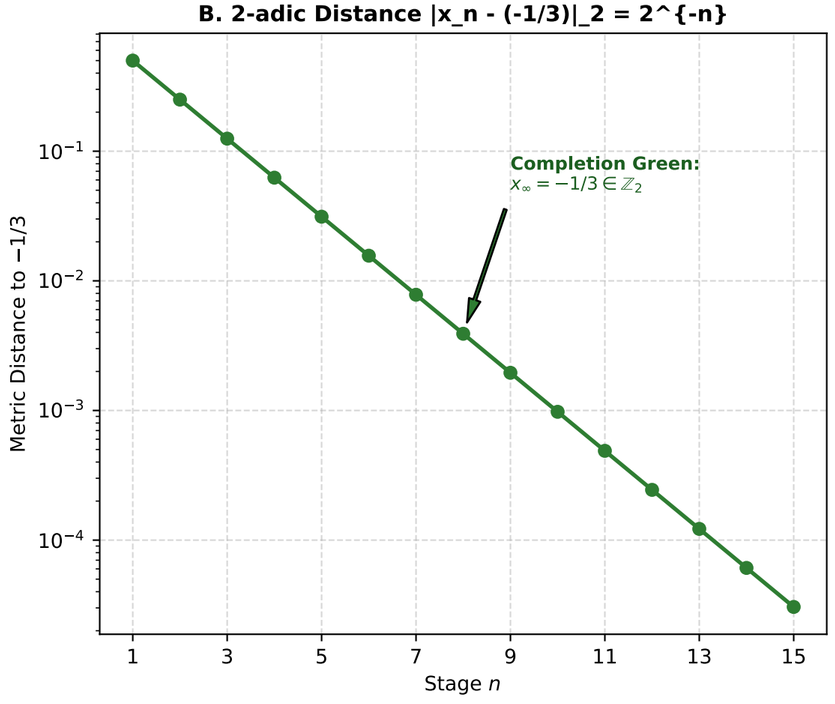
Local Green, Global Red IV: Finite-Stage Solvability and Completion-Relative Obstruction





C. Carrier Reflection: $\eta : \mathbb{Z} \hookrightarrow \mathbb{Z}_2$

Original Carrier $\mathbb{Z}$ (Integers)

Global Red: $\forall x \in \mathbb{Z},\ \exists n : 3x + 1 \equiv 0 \pmod{2^n}$
No solution in $\mathbb{Z}$ because $-1/3 \notin \mathbb{Z}$

Canonical Reflection η

Completion $\mathbb{Z}_2$ (2-adic Integers)

Unique Solution: $x_\infty = -1/3 \in \mathbb{Z}_2$
Residual: $-1/3 \notin \mathrm{im}(\eta) \Rightarrow [x_\infty] \neq 0 \in \mathbb{Z}_2/\mathbb{Z}$

Completion Green does NOT promote to Original Green